

One-photon attention notation

$$x_i \in \mathbb{R}^d \longrightarrow z_i = V x_i$$

$$s_{ij} = \langle W x_i, W x_j \rangle$$

$$A_{ij} = \text{softmax}_j(s_{ij})$$

B: output uses $A_{ij} z_j$ **C:** output uses $V(A_{ij} x_j)$

Compound-sector notation

$$\mathcal{M} = \mathcal{M}_{\text{patch}} \oplus \mathcal{M}_{\text{feat}}$$

$$\Omega_{\times} = (1_{\text{patch}}, 1_{\text{feat}})$$

Ω_{pp}, Ω_{ff} retain same-block 2-photon sectors

D: keep $\Omega_{\times}$ only **D_full:** keep $\Omega_{\times} \cup \Omega_{pp} \cup \Omega_{ff}$